

Mingxi Yan

PhD candidate in Computer Science at Georgia State University. Simulation, data assimilation, and machine learning for complex dynamical systems.

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Education

Ph.D. in Computer Science 2023 – expected 2027

Georgia State University, Atlanta, GA. Advisor: Prof. Xiaolin Hu

M.S. in Computer Science 2020 – 2022

Georgia State University, Atlanta, GA

B.S. in Computer Science 2014 – 2018

NingboTech University, Ningbo, China

Research

Learned observation operators for data assimilation 2023 – present

Dissertation research

- Learns the observation operator inside a particle filter with a neural network, in place of a hand-written formula. Target: particle degeneracy, the main failure mode of particle filters.
- Trained on real Caltrans PeMS highway sensor data; tested on another year and other sensors.
- Lower error than a fully calibrated physics-based baseline in free-flowing traffic and overall. Congested traffic remains the hard case.
- The same trained network runs inside an ensemble Kalman filter with no changes.
- Built the full stack alone: data pipeline, SUMO traffic simulation, PyTorch training, sequential Monte Carlo inference, and evaluation. Manuscript in preparation.

Online calibration of simulation models

With Prof. Xiaolin Hu. SIMULATION 2024; Winter Simulation Conference 2024

- Particle filters that estimate a discrete-event simulation's state and its model parameters together as observations arrive, so the model is calibrated while it runs.
- With several parameters, different combinations can fit the same observations. The WSC 2024 paper studies that case. Experiments use synthetic data (identical-twin setup).

Wildfire simulation on the web 2020 – present

Systems Integrated Modeling and Simulation Lab, Georgia State University

- Independently built the front end and back end of FireSim, a map-based web app for planning prescribed burns: outline a land unit, set the ignition plan, run the fire simulation. Live at firesim.cs.gsu.edu. First-author paper at IEEE SoutheastCon 2023.
- Built the service and web layer of the lab's cloud wildfire simulation tools: API wrappers, front-end and back-end framework, data processing, and web display.
- Other students in the lab now build on this framework.

Experience

Graduate Research Assistant Aug 2020 – present

Systems Integrated Modeling and Simulation Lab, Georgia State University, Atlanta, GA

Graduate Teaching Assistant Fall 2026

Department of Computer Science, Georgia State University

Lead Lecturer, Informatics Olympiad (C/C++) Aug 2019 – Jul 2020

Tarena (Tongcheng Tongmei), Beijing, China

- Designed the curriculum and taught algorithms and problem solving to students training for national informatics olympiads.

Development Engineer Jun 2018 – Jul 2019

Zhejiang Changatang Network Co., Ltd., Hangzhou, China

- Backend development in C#; designed and maintained the databases behind the services.

Publications

- [1] Xiaolin Hu and **Mingxi Yan**. Data assimilation for online model calibration in discrete event simulation. *SIMULATION* 100(6), pp. 529–544, 2024. doi:10.1177/00375497231221578
- [2] Xiaolin Hu and **Mingxi Yan**. Data Assimilation for Online Calibration of Simulation Digital Twin - A Case Study With Multiple Model Parameters. *Proceedings of the 2024 Winter Simulation Conference (WSC)*, pp. 2844–2855. doi:10.1109/WSC63780.2024.10838855
- [3] Xiaolin Hu, **Mingxi Yan**, Tony Derado, Wei Zhao, Bernard Zeigler, Doohwan Kim, and Chungman Seo. WIP: Towards Cloud-based Wildland Fire Simulation Service. *2024 IEEE International Conference on Service-Oriented System Engineering (SOSE)*, pp. 20–24. doi:10.1109/SOSE62363.2024.00009
- [4] **Mingxi Yan** and Xiaolin Hu. Towards A Map-Based Web Application for Prescribed Fire Simulation. *IEEE SoutheastCon 2023*, pp. 920–926. doi:10.1109/SoutheastCon51012.2023.10115091

Skills

Methods: data assimilation, particle filters (sequential Monte Carlo), ensemble Kalman filters, state-space models, uncertainty quantification, deep learning, discrete-event and traffic simulation.

Programming: Python, PyTorch, C/C++, C#, Java, TypeScript.

Tools: SUMO, React, Node.js, SQLite, Git, Linux.

Languages: Chinese (native), English (fluent).